



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

AWS STUDENT BUILDER GROUP

Wave 'n' Wire: Edge-to-Cloud Workshop

// From Signals to Smart Systems //

60 STUDENTS · 2 DAYS · BEGINNER-FRIENDLY · 22-23 OCTOBER

Welcome to Wave 'n' Wire

DAY 1 – AWS FOUNDATIONS & CLOUD COMMUNICATION

DAY 2 – AI, IOT & SYSTEM INTEGRATION

PRACTICAL EDGE-TO-CLOUD EXPOSURE

PREREQUISITES

Before You Arrive

- No prior cloud-computing experience is required.
- Basic familiarity with electronics and computer usage is helpful.
- Bring a laptop with a working internet connection and a modern web browser.
- Have access to an email account for the required learning platforms and AWS activities.
- Be prepared to create and test simple cloud-based projects.

WORKSHOP BRIEF

BEGINNER-FRIENDLY // 2-DAY WORKSHOP // HANDS-ON

Wave 'n' Wire: Edge-to-Cloud Workshop introduces students to cloud computing, essential AWS services, AWS IoT Core, and MQTT-based communication. Participants learn how data moves from an edge input to a cloud service and produces a corresponding system response, building the technical foundation needed for the upcoming Wave 'n' Wire event without revealing the complete competition workflow.

DAY 1 – AWS FOUNDATIONS AND CLOUD COMMUNICATION

SESSION 1

Introduction to Cloud Computing and AWS

- Introduction to cloud computing, AWS, and the role of cloud services in modern applications.
- Walkthrough of the AWS Management Console and commonly used AWS services.

Checkpoint: understands cloud computing & major AWS services.

SESSION 2

Overview of Core AWS Services

- Amazon EC2, S3, Lambda, DynamoDB, API Gateway, and AWS IoT Core.
- Focus on the purpose and practical use of each service.

Checkpoint: can identify suitable AWS services for a use case.

SESSION 3

AWS IoT Core and MQTT Fundamentals

- Introduction to IoT and MQTT communication.
- Publish/subscribe architecture, brokers, topics, and payloads.
- AWS IoT things, certificates, and policies.

Checkpoint: understands MQTT topics & pub/sub communication.

SESSION 4

Guided MQTT Hands-on Activity

- Connect an MQTT client and subscribe to a topic.
- Publish messages, modify payloads, observe device-state messages.

Checkpoint: publishes & receives a test message.

SESSION 5

Broad Introduction to the Main Event

- Central idea of Wave 'n' Wire; relationship between AI, electronics, cloud, and virtual smart environments.
- Detailed competition workflow not disclosed at this stage.

Checkpoint: understands the broad concept & relevance of the event.

SESSION 6

Assignment Instructions

- Introduction to the Day 1 assignment.
- Submission expectations and basic instructions.

Checkpoint: understands assignment & submission process.

DAY 2 – AI, IOT AND SYSTEM INTEGRATION

SESSION 1

Quick Revision and Setup

- Short revision of AWS IoT Core, MQTT topics, and pub/sub communication.
- Recap of message payloads.

SESSION 2

AWS IoT Core Hands-on Practice

- Practise connecting to AWS IoT Core.
- Publish/receive messages via meaningful topics; modify payloads.

SESSION 3

Introduction to Teachable Machine

- Create a simple image or gesture-recognition model.
- Collect sample inputs, train the model, and test its output.

Checkpoint: creates & tests a basic recognition model.

SESSION 4

Guided Edge-to-Cloud Integration Activity

- Combine AI recognition with MQTT communication in a demonstration workflow.
- Flow: recognition → message → MQTT → simulated response.

Checkpoint: demonstrates recognition-to-MQTT-to-response flow.

SESSION 5

Connecting Workshop Skills to the Main Event

- Recap of how AI, electronics, MQTT, and IoT Core connect.
- How virtual-system interaction supports the main event.

SESSION 6

Assignment Results and Closing

- Assignment result announcement.
- Final doubts and workshop closing.

KEY TAKEAWAYS

What You'll Walk Away With

- Understand the fundamentals of cloud computing and AWS.
- Identify the purpose of major AWS services.
- Understand the role of AWS IoT Core in connected systems.
- Use MQTT topics, messages, publishing, and subscribing.
- Gain basic familiarity with AWS IoT Core connections and authentication concepts.
- Create and test a simple AI-based recognition model using Teachable Machine.
- Understand how edge inputs can be connected to cloud-based actions.
- Gain practical exposure to an edge-to-cloud workflow.
- Be better prepared to use AI, IoT, and cloud communication.

CERTIFICATE OF PARTICIPATION

LEARNING RESOURCES

WORKSHOP NOTES

CODE EXAMPLES ACCESS

CERTIFICATE OF EXCELLENCE

READY TO BUILD ACROSS THE EDGE-TO-CLOUD WORKFLOW

Workshop Guide

Wave 'n' Wire: Edge-to-Cloud Workshop // Department of Electronics and Communication Engineering // 22-23 October

SESSION	DAY	TOPIC	OUTCOME
D1 · S1	Day 1	Introduction to Cloud Computing and AWS	Understands cloud & major AWS services
D1 · S2	Day 1	Overview of Core AWS Services	Can identify suitable AWS services
D1 · S3	Day 1	AWS IoT Core and MQTT Fundamentals	Understands MQTT topics & pub/sub
D1 · S4	Day 1	Guided MQTT Hands-on Activity	Publishes & receives a test message
D1 · S5	Day 1	Broad Introduction to the Main Event	Understands the event's broad concept
D1 · S6	Day 1	Assignment Instructions	Understands assignment process
D2 · S1	Day 2	Quick Revision and Setup	Refreshed on IoT Core & MQTT
D2 · S2	Day 2	AWS IoT Core Hands-on Practice	Comfortable with IoT Core pub/sub
D2 · S3	Day 2	Introduction to Teachable Machine	Creates & tests a recognition model
D2 · S4	Day 2	Guided Edge-to-Cloud Integration Activity	Demonstrates full recognition-to-response flow
D2 · S5	Day 2	Connecting Workshop Skills to the Main Event	Connects skills to event context
D2 · S6	Day 2	Assignment Results and Closing	Results shared; workshop closed

Wave 'n' Wire: Edge-to-Cloud Workshop

A two-day, beginner-friendly workshop that takes participants from the fundamentals of cloud computing through AWS IoT Core, MQTT communication, and AI-based recognition with Teachable Machine — closing with a guided activity that connects an edge input to a cloud-driven response.

QUICK OVERVIEW – THE FLOW

60

STUDENTS

2

DAYS

12

SESSIONS

Cert.

ISSUED

Key Takeaways

Understand the fundamentals of cloud computing and AWS.

Identify the purpose of major AWS services.

Understand the role of AWS IoT Core in connected systems.

Use MQTT topics, messages, publishing, and subscribing.

Gain basic familiarity with AWS IoT Core connections and authentication concepts.

Create and test a simple AI-based recognition model using Teachable Machine.

Understand how edge inputs can be connected to cloud-based actions.

Gain practical exposure to an edge-to-cloud workflow.

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